Section Id :	64065330372
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	2
Number of Questions to be attempted :	2
Section Marks :	25
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065367932
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 1 Question Id : 640653470753 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531564240. ✔ YES

6406531564241. ✖ NO

Sub-Section Number : 2

Sub-Section Id : 64065367933

Question Shuffling Allowed : No

Is Section Default? : null

Question Number : 2 Question Id : 640653470754 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 25

Question Label : Multiple Choice Question

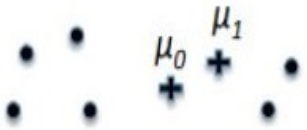
1. (1 point) In K-means, the number of clusters (K) changes as the algorithm reaches to convergence.
   A. True
   B. False

2. (1 point) A uni-variate Gaussian distribution is completely defined by the following parameters.
   A. mean, variance
   B. median, covariance
   C. mode, variance
   D. mode, covariance

3. (1 point) K-Means will always find the global minimum.
   A. True
   B. False

4. (3 points) Consider the set of training data below, and two clustering algorithms: K-Means and a Gaussian Mixture Model (GMM) trained using EM. Will these two clustering algorithms produce the same cluster centers (means) for this data set? Explain why or why not.

5. (3 points) Assume there are two Gaussian components in the GMM; μ_0, μ_1, σ_0 and σ_1 define means and variances of these two components, π_0 and $(1-\pi_0)$ denote the mixture proportions of the two Gaussians (i.e. $p(x) = \pi_0 N(\mu_1, \sigma_1) + (1-\pi_0)N(\mu_2, \sigma_2)$).



(a) (1 point) Draw on the figure the directions in which μ_0 and μ_1 will move during the next M-step.

(b) (2 points) Will the estimate of π_0 increase or decrease on the next EM step? Explain your reasoning in one sentence.

6. (5 points) We saw two approaches to solve the sequence prediction problem: greedy and viterbi. Consider a weather sequence prediction problem, assuming first-order markov chain. Given that Sunny weather was observed on Day-0, compute the best possible weather forcast sequence for the next three days using 1) greedy approach, 2) viterbi approach.

Hint: For viterbi keep a track of path taken to calculate $\gamma_A(B)$, where $\gamma_A(B)$ denotes maximum probability of reaching state A at time step B.

Yesterday \ Today	Sunny	Cloudy	Rainy
Sunny	0.5	0.375	0.125
Cloudy	0.25	0.125	0.625
Rainy	0.25	0.075	0.675

7. (3 points) We saw two approaches for vector quantization, namely: k-means & Linde-Buzo-Gray Algorithm (LBG). Explain in detail the clustering mechanism used in these two algorithms, highlighting differences in their approach.

8. (1 point) For perfect reconstruction of a bandlimited signal, the sampling frequency must be ____.

   A. equal to twice the maximum frequency

   B. greater than equal to twice the maximum frequency

   C. lesser than equal to twice the maximum frequency

   D. none of the above

9. (3 points) For each of the following discrete time system, comment upon the following system properties (1) linear/non-linear, (2) time-invariant/time-variant, and (3) causal/anti-causal.

   (a) (1.5 points) $y[n] = nx[n]$

   (b) (1.5 points) $y[n] = e^{x[n]}$

10. (1 point) Any point in a 3-D space can be represented as a ____ combination of its basis functions.

11. (1 point) Let $f_X(x)$ be the probability distribution function for random variable X then

$$\int_{-\infty}^{\infty} f_X(x)dx = ?$$

   A. 1

   B. ∞

   C. 0

   D. none of the above

12. (1 point) The Gaussian curve is always symmetrical about ____.

   A. 1

   B. 0

   C. standard deviation

   D. mean

13. (1 point) Psychophysical equivalent of frequency is ____.

Options :

6406531564242. ✔ I have written answers on the answer sheets

6406531564243. ✖ Not applicable

Section Id :	64065330373
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	20
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065367934
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 3 Question Id : 640653470755 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0"

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Options :

6406531564244. ✔ YES